

Evidence pdf

These pictures and test texts are the part for which I prompt Gemin ai for helping me to do correctly what is asked. I leaned so many things.

Welcome to the course! Getting your environment set up can feel like a lot of moving parts, but we will break this down step-by-step so you can complete your lab seamlessly.

Since your lab is due today (4 June 2026), let's get you set up and coding right away. Follow this sequence exactly, and you'll have everything done and ready for submission.

Step 1: Install Your Tools (Where to click & what to download)

Before writing code, you need the software installed on your physical computer.

1 Install the Code Editor (VS Code):

- Go to code.visualstudio.com.
- Download and run the installer for your OS (Windows, Mac, or Linux).
- **CRITICAL FOR WINDOWS:** During installation, look out for a checkbox that says "Add to PATH" and make sure it is **checked**.

2 Install Python:

- Go to python.org/downloads.
- Download Python 3.10 or 3.11.
- **CRITICAL FOR WINDOWS:** On the very first screen of the installer, check the box at the bottom that says "Add Python to PATH" or "Add python.exe to PATH" before clicking install.

3 Install Git:

- **Windows:** Download and install from git-scm.com.
- **Mac:** Open your built-in Terminal app and type `brew install git` (or it will auto-prompt you to install Xcode Command Line Tools when you type `git` into the terminal).

Step 2: Set Up Your Project Folder & Virtual Environment

Now, let's create a dedicated folder for your work and configure your isolated Python workspace.

- 1 Open VS Code.
- 2 Open the built-in terminal inside VS Code by pressing `Ctrl + ``` (backtick) or going to Terminal

- 1 Open VS Code.
- 2 Open the built-in terminal inside VS Code by pressing `Ctrl + ``` (backtick) or going to **Terminal > New Terminal** in the top menu.
- 3 Copy and paste the following commands into that terminal one by one to create your project directory and virtual environment:

For Windows:

Bash

```
# Create the folder
mkdir %USERPROFILE%\ai_labs
cd %USERPROFILE%\ai_labs

# Create the environment
python -m venv ai_env

# Activate it
```

Step 5: Complete Exercise 2 (Git & GitHub Setup)

1. Go to github.com and log in (or create an account using your university email).
2. Click **New Repository**. Name it ai-course-lab0. Leave it public, do **not** add a README or .gitignore yet (we will push your local files instead). Click **Create repository**.
3. Copy the URL of your new repository (it looks like `https://github.com/your-username/ai-course-lab0.git`).
4. Go back to your VS Code terminal and configure your identity (replace with your actual details):

Bash

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your.email@ashesi.edu.gh"
```

```
git config --global init.defaultBranch main
```

5. Initialize Git and link it to GitHub:

Bash

```
git init
```

```
git remote add origin https://github.com/your-username/ai-course-lab0.git
```

6. Create a .gitignore file so you don't upload giant system files. Create a new file named .gitignore and paste this inside:

Plaintext

```
__pycache__/
```

```
*.pyc
```

```
.DS_Store
```

```
ai_env/
```

```
.ipynb_checkpoints/
```

7. Create a README.md file and type out your course name, semester, your name, and your Student ID.
8. Commit and Push your work:

Bash

```
git add .
```

```
git commit -m "Initial commit with environment checks"
```

```
git branch -M main
```

```
git push -u origin main
```

(GitHub may prompt you to log in via a browser window to authenticate the push).

(Submission Item #2 Complete!)

Step 6: Complete Exercise 3 (Google Colab Practice)

1. Open your browser and go to colab.research.google.com.
2. Click **New Notebook**.
3. Change the title at the top left from Untitled0.ipynb to colab_practice.ipynb.
4. Create a Text cell at the top by clicking **+ Text** and type: # Lab 0: Colab Practice - [Your Name].

5. Look at the top right menu of Colab, click the down arrow next to "Connect", choose **Change runtime type**, select **T4 GPU**, and hit **Save**.
6. Copy the code block from **Exercise 3** in your assignment instructions into separate code cells in Colab and click the "Play" button on each cell to run them.
7. Once everything runs successfully, go to **File > Download > Download .ipynb**. Save this file to your computer. **(Submission Item #3 Complete!)**

Step 7: Complete Graded Exercises 4 to 7 (The Coding Phase)

Now we will build your primary lab submission file.

1. Back in VS Code, open the command palette (Ctrl+Shift+P or Cmd+Shift+P) and type Jupyter: Create New Blank Notebook.
2. Save this file immediately as lab0_exercises. ipynb.
3. You will create individual code cells for Exercises 4, 5, 6, and 7. Below are the complete, functional blocks of code to paste into those cells to solve the tasks requested: